Parv **Kapoor**

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Education

Carnegie Mellon University *Pittsburgh, PA*
Ph.D. in Software Engineering | Advisors: Dr. Eunsuk Kang and Dr. Sebastian Scherer *August 2026 (expected)*

Carnegie Mellon University *Pittsburgh, PA*
Masters in Software Engineering | GPA: 4.03/4.00 *December 2025 (expected)*

Manipal Institute of Technology *Manipal, India*
B.Tech. in Computer Science Engineering | GPA: 8.59/10 *August 2020*

Research Areas: Foundation Models, Embodied AI, Software Engineering for Intelligent Systems

Skills

Programming Python, C, C++, CUDA programming, JAVA, MATLAB, Alloy, TLA+, MySQL, OpenCL
Tools and Libraries PyTorch, TensorFlow, ROS, Docker, Habitat, Isaac Gym

Ph.D. Research

AirLab, *Carnegie Mellon University* *Pittsburgh, PA*
PI: Sebastian Scherer *August 2021 – Ongoing*

- Built constrained decoding frameworks for robotic transformers using formal logic over token spaces, outperforming baselines by 25%
- Pretrained LfD policies and refined them online via symbolic tree search, improving constraint satisfaction by 60% [ICRA'23]
- Developed vision-based control barrier functions for UAV safety, validated via sim-to-real transfer from Isaac Sim to field, achieving 71% improvement over baselines across 70+ flight hours [ICRA'22, RSS'25]

Software Design and Analysis Lab, *Carnegie Mellon University* *Pittsburgh, PA*
PI: Eunsuk Kang *August 2021 – Ongoing*

- Built STLCG++, a PyTorch/JAX library for differentiable STL, enabling 100× faster robustness computation via attention masking [RA-L]
- Trained world models and used pretrained visual embeddings (DINOv2, CLIP) with MPPI for logic-constrained object navigation
- Learned formal behavior rules from robot demonstrations using MaxSAT for structured policy synthesis [ICSE'25]
- Designed stochastic optimization methods to analyze RL policy robustness under environmental deviations [FM'24]
- Devised requirement decomposition for STL-constrained trajectory optimization, reducing solve time by 65% [NFM'24]

Experience

Microsoft Research *New York, U.S.A.*
Machine Learning Research Intern | PI: Dr. John Langford and Dr. Siddhartha Sen *May 2025 – Ongoing*

- Researching matryoshka representation learning for belief state transformers for efficient long-horizon reasoning with large language models

Scaled Foundations *Seattle, WA*
Student Researcher (Part time) | PIs: Dr. Jonathan Huang and Dr. Ashish Kapoor *September 2024 - May 2025*

- Developed pipelines for distributed synthetic data generation, pretraining and fine tuning of robot foundation models
- Developed aligned robotics foundation models using novel fine tuning paradigms for navigation and instruction following
- Contributing to the development of the AirGen simulation platform for aerial, ground, and manipulator robots

Scaled Foundations *Seattle, WA*
Research Intern | PIs: Dr. Sai Vemprala and Dr. Ashish Kapoor *May 2024 - August 24*

- Developed a novel technique for safe trajectory generation extending state-of-the-art autoregressive robotics pretraining architecture, outperforming baselines by 74.3%.
- Built a foundation model field deployment pipeline for diverse robots (Unitree Go2, DJI Robomaster) using domain adaptation.
- Integrated multimodal foundation models into the GRID platform, developed and deployed skills chaining the models.

Verimag, *Université Grenoble Alpes* *Grenoble, France (Remote)*
Research Engineer | PI: Dr. Thao Dang *January 2021 - August 2021*

- Developed input stimulus generation theory using timed automata for autonomous system validation
- Evaluated these techniques within the SUMO simulation environment for applications in autonomous vehicles

Cyber Physical Systems Lab, *University of Southern California* *Los Angeles, U.S.A.*

Research Intern | PI: Dr. Jyotirmoy Vinay Deshmukh *January 2020 - January 2021*

- Developed novel model-based reinforcement learning algorithms for safe policy training from signal temporal logic specifications
- Implemented efficient model-free algorithms (TRPO, A3C, PPO) in PyTorch with unique STL-based reward design
- Achieved 82% higher specification satisfaction compared to baseline RL policies
- Engineered in-house simulation environments for algorithm benchmarking employing CARLA, AirSim, and Gazebo

Visual Computing Group, *Cardiff University* *Cardiff, U.K.*

Research Intern | PI: Dr. David Marshall *May 2019 - July 2019*

- Constructed a safe trajectory prediction system for visually impaired individuals using ZED stereo camera
- Implemented and trained a 2-stream CNN in TensorFlow on human walking data for forecasting ego agent camera movement
- Improved CNN accuracy in low-data regimes through neuro-inspired data augmentation

RapidQube Digital solutions Pvt. Ltd. *Mumbai, India*

Research Intern *May 2018 - July 2018*

- Created an accident prediction system leveraging convolutional neural networks and object tracking algorithms (YOLOv3)
- Implemented depth prediction Residual CNNs alongside YOLO v3 in Tensorflow to classify nearby drivers' speed profiles with 300 ms latency

Preprints and Publications

Constrained Decoding for Robotics Foundation Models

P. Kapoor, A. Ganlath, C. Liu, S. Scherer, E. Kang

- Under Submission

Pretrained Embeddings as a Behavior Specification Mechanism

P. Kapoor, A. Hammer, A. Kapoor, K. Leung, E. Kang

- Under Submission [arxiv]

ViSafe: Vision-enabled Safety for High-speed Detect and Avoid

P. Kapoor, I. Higgins, N. V. Keetha, J. Patrikar I. Cisneros, Z. Ye, Y. He, Y. Hu, C. Liu, E. Kang, S. Scherer

- **Robotics Science and Systems (RSS)** 2025 [arxiv]

STLCG++: A Masking Approach for Differentiable Signal Temporal Logic Specification

P. Kapoor, K. Mizuta, E. Kang, K. Leung

- **IEEE Robotics and Automation Letters (RA-L)**, presented at IEEE International Conference on Robotics and Automation (ICRA) 2026 [arxiv]

Logically Constrained Robotics Transformers for Enhanced Perception Action Planning

P. Kapoor, S. Vemprala, A. Kapoor

- **Robotics Science and Systems (RSS): Towards Safe Autonomy** 2024 [arxiv]

Constrained LTL Specification Learning from Examples

C. Zhang*, **P. Kapoor***, I. Dardik, A. Cui, R.M. Goes, D. Garlan, E. Kang

- **IEEE International Conference on Software Engineering (ICSE)** 2025 [arxiv]

Analyzing Tolerance of Reinforcement Learning Controllers against Deviations in Cyber Physical Systems

C. Zhang*, **P. Kapoor***, R.M. Goes, D. Garlan, E. Kang, A. Ganlath, S. Mishra, N. Ammar

- **Formal Methods (FM)** 2024 [arxiv]

Safe Planning through Incremental Decomposition of Signal Temporal Logic

P. Kapoor, R.M. Goes and E. Kang

- **Nasa Formal Methods (NFM)** 2024 [arxiv]

Follow The Rules: Online Signal Temporal Logic Tree Search for Guided Imitation Learning in Stochastic Domains

J. Patrikar, J. Aloor, **P. Kapoor**, S. Scherer and J. Oh

- **IEEE International Conference on Robotics and Automation (ICRA)** 2023 [arxiv]

Challenges in Close-Proximity Safe and Seamless Operation of Manned and Unmanned Aircraft in Shared Airspace

J. Patrikar, J. Dantas, S. Ghosh, **P. Kapoor** et al

- **IEEE International Conference on Robotics and Automation (ICRA)** 2022 [arxiv]

Model-based Reinforcement Learning from Signal Temporal Logic Specifications

P. Kapoor, A. Balakrishnan, J. V. Deshmukh

- [arxiv]